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MICROPROCESSORS AND MICROCONTROLLERS
Course Project | Abstract

AeroLab Pro: Comprehensive IoT-Enabled Drone Environment Simulator and Telemetry Dashboard

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Abstract

Traditional aeronautical and drone development requires expensive flight hardware and carries a significant risk of catastrophic crashes during early algorithmic testing. AeroLab Pro addresses this challenge by providing a comprehensive, breadboard-mounted hardware-in-the-loop simulator that physically replicates a drone's sensory environment. By physically manipulating the board—tilting, rotating, and introducing obstacles—users can simulate real-world flight dynamics without a physical airframe. The system acts as an educational and testing bridge, demonstrating the practical application of embedded protocols (I^2C, ADC, PWM, UART) mapped directly to the microprocessors syllabus.

The hardware architecture leverages an ESP32 microcontroller as the central flight computer, interfacing with a diverse 10-sensor cluster. Spatial attitude (roll, pitch, yaw) is captured by an MPU6050 IMU using Kalman filtering, while altitude and environmental obstacles are measured using dual HC-SR04 ultrasonic sensors, one of which is mounted to an SG90 servo for a 180-degree radar sweep. Environmental health is monitored via a BMP180 (barometric pressure), DHT11 (temperature/humidity), MQ FC-22 (gas/smoke), and an LDR (ambient light for night-mode activation). For standalone operation without a laptop, local feedback is provided through an OLED display, an audible buzzer, and a state-indicating RGB LED.

Crucial to the simulator's capability is its real-time telemetry streaming over WiFi using the WebSocket protocol. All sensor telemetry is broadcast at 25Hz to a browser-hosted "glass cockpit" dashboard. This interface features a live 3D drone model that perfectly mirrors the physical board's orientation, alongside a Primary Flight Display (PFD) presenting an artificial horizon. The dashboard also renders a 2D sonar radar map from the servo sweep, dynamic gas sparklines, vibration intensity bars, and a timestamped flight event log.

The system was successfully validated through interactive physical testing: board manipulation accurately updated the 3D model, proximity to the front sensor triggered cascading visual and auditory alerts, and sensor interference produced immediate dashboard responses. Ultimately, AeroLab Pro successfully demonstrates complex sensor fusion, real-time wireless communication, and advanced web-based data visualization, delivering a complete drone telemetry experience within a low-cost, static embedded platform.

Keywords: ESP32, MPU6050, Kalman Filter, Telemetry Dashboard, WebSocket, Sensor Fusion, Hardware-in-the-Loop, Primary Flight Display, Radar Sweep.